

Mark Raymond Takla

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Dual American-Egyptian Nationality | Military Exempted

TECHNICAL SKILLS AND LANGUAGES

Languages: Java, Python, C, C#, SQL, MIPS Assembly, Haskell, Prolog, R, HTML/CSS. Frameworks & Libraries: Next.js, React, ASP.NET, FastAPI, LangChain, LangGraph, Scikit-Learn, TensorFlow, Keras, Pandas, NumPy.

Software & Tools: UIPath (RPA), PowerBI, MATLAB, OrCAD Pspice, Adobe Photoshop, AfterEffects, Microsoft Office.

Platforms: Linux, Git, SQL Server, MongoDB, SQLite, AutoCAD.

Spoken Languages: Arabic & English (Proficient), German (A2), French (Basics).

EDUCATION

German University in Cairo (GUC)

Bachelor of Engineering | Major in Computer Science

09/2023 - Current

Cumulative German Scale GPA: 2.15 (equivalent to 3.3 (B+) American Scale GPA)

Cairo English School (CES) and New Ramses College (NRC) (IGCSE System)

Earlier Years

Early graduation after year 11 | Final score 110% with scale factor

WORK EXPERIENCE

German University in Cairo

Computer Science Junior Teaching Assistant

09/2024 – 06/2026

- CSEN102/CSIS102 (Python): Instructed 60+ students in core programming and data structures.
- CSEN401 (OOP Java): Mentored 50+ students and held 2 weekly project office hours to guide complex software projects.
- Gained many soft-skills such as leadership, communication, teaching, patience, accountability, and time management.

Bank NXT

Robotic Process Automation (RPA) Internship

07/2025 – 08/2025

- Completed a multidisciplinary rotation across the banking industry, including intensive weeks in RPA, Integration and IT Governance & Risk Management.
- Developed 5+ unattended UIPath robots for automated data entry and web scraping, saving significant bank resources and eliminating manual form filling time by 100%. ([Demo Link](#))

GUC Innovators and GUC Google Developer Groups

Vehicle Dynamics Engineer in Innovators and Robotics Engineer in GUC GDG

06/2024 – 07/2025

Alstom

Rotational Engineering Internship

06/2024 – 08/2024

- Completed a comprehensive rotation through V&V, ICC, CBI, and RCM departments
- Performed Verification & Validation (V&V) applying ISTQB principles and SCADA systems to analyze Automatic Train Control and ensure (ATC/ATS) mission-critical reliability.
- Identified 4 technical discrepancies in signaling test cases within the HMI to ensure accurate train simulations for the Turkey UAG line.

PROJECTS (Full List (25+) on [GitHub](#) and [Portfolio Website](#))

University HR Management System ([GitHub Repository](#))

12/2025

- Engineered a comprehensive relational database using SQL Server DBMS (including EERD and relational schema) with 40+ stored procedures to drive a responsive ASP.NET & JS fully functional website for multi-rank data management and automated payroll logic.

Customer churn benchmarking on two tabular datasets ([GitHub Repository](#))

03/2026

- Modeled churn using Scikit-learn pipelines (StandardScaler, OneHotEncoder), GridSearchCV, and joblib serialization; benchmarked Random Forests and decision trees against tuned Logistic Regression using Matplotlib/Seaborn confusion-matrix visualizations.

Production-Ready Email AI Assistant ([GitHub Repository](#))

04/2026

- Built an automated email assistant using LangGraph Cloud, featuring a React-based Chat UI that supports Human-in-the-loop interrupts and persistent thread management.

Jackaroo Computer Game with GUI ([GitHub Repository](#))

05/2025

- Developed a multiplayer card game using JavaFX and Java 23, applying the MVC architecture for clean, modular design.

Advanced Persistent RDBMS Engine ([GitHub Repository](#))

05/2026

- Engineered a high-performance engine using a 3-tier hierarchy (DBMS, Table, Page) with disk-based serialization and O(1) pointer-based record access.

Certificates, Competitions and Volunteering ([Full list of 11+ Certificates on Portfolio Website](#))

- **Certifications:** *Data Science & AI Diploma* (Sprints.ai - 250hrs and Space Code 180hrs); *Intro to LangChain* (LCA); *AI Data Analytics Bootcamp* (EYouth); *Data Analysis Using R* (SRTA-City).
- **Community:** Scout Leader (2013–Present); Managed summer camps in New Jersey, USA for 30+ children; Organized charity activities and ceremonies for 200+ Sudanese-Egyptian refugee children.